

MINDCRAFT
HACKATHON TEAM

SMART INDIA HACKATHON 2026



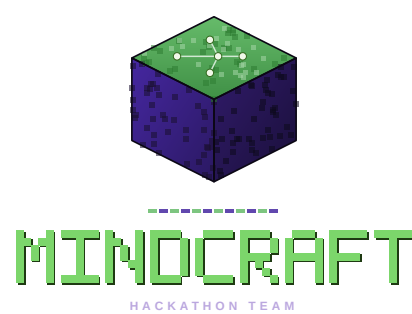
SMART INDIA
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TITLE PAGE

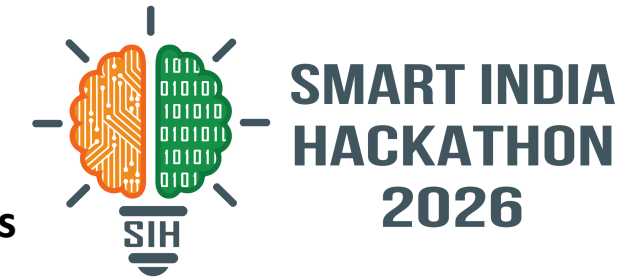
- **Problem Statement ID** – SIH26167
- **Problem Statement Title** SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-** 134616
- **Team Name** - MINDCRAFT



IDEA TITLE



SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

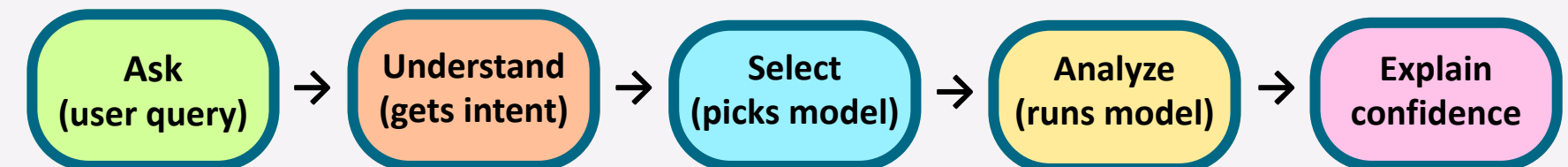


IDEA / SOLUTION

SatQuery AI: Conversational Satellite Analysis

- **Query Naturally:** Ask questions in plain language no GIS expertise needed.
- **Track Time:** Compare past and present imagery to detect critical changes.
- **Fuse Modalities:** Combine Optical & SAR data for all-weather, deep insights.
- **See the Proof:** Get answers backed by visually highlighted bounding boxes and a confidence score.
- **Global Access:** Multilingual support breaks down language barriers. (22 Regional Indian Languages Supported)

HOW IT ADDRESSES THE PROBLEM

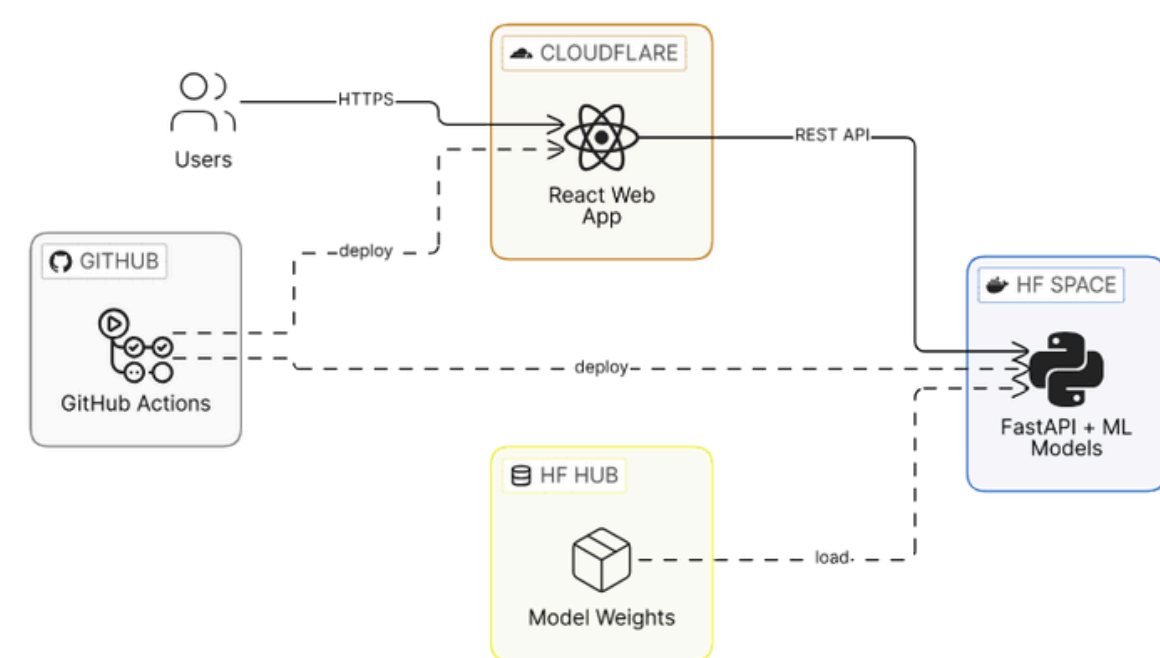


Existing Gaps | Our Approach

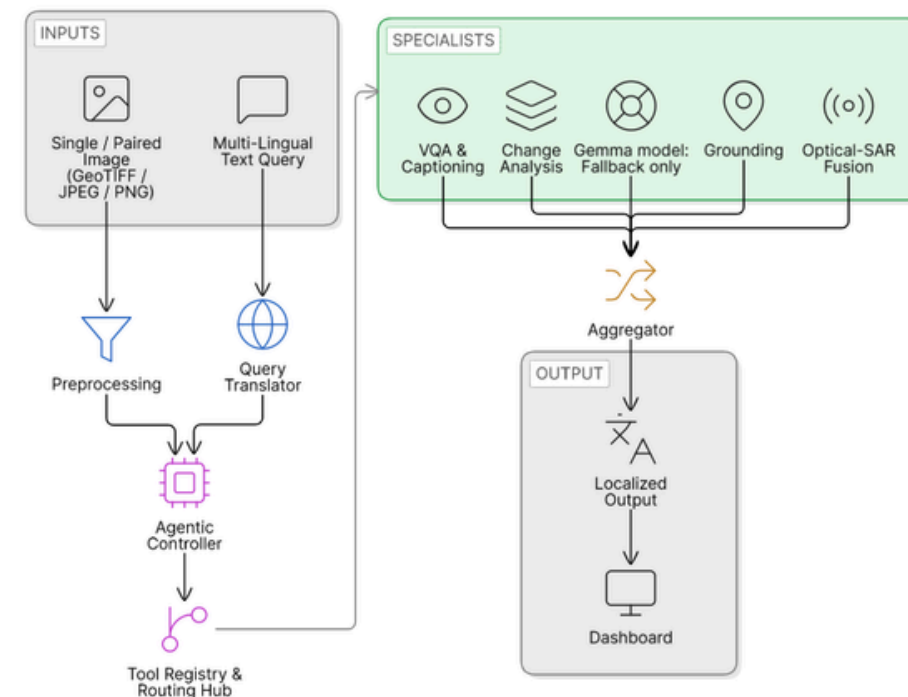
- Requires GIS skills | Chat in plain language
- Single-task models | Rule-based task routing
- Unverified answers | Visually grounded evidence
- Generic LLMs hallucinate | Grounded answers with confidence scores

INNOVATION & UNIQUENESS

- **From pixels to conversations:** Chat directly with imagery instead of manual GIS interpretation.
- **One question, multiple perspectives:** Fuses optical, SAR and temporal data, in line with India's multi-sensor EO goals.
- **Evidence, not assumptions:** Every answer is visually grounded with highlights on the image.
- **Inclusive Intelligence:** IndicTrans2 translates queries and answers across 22 Indian languages.

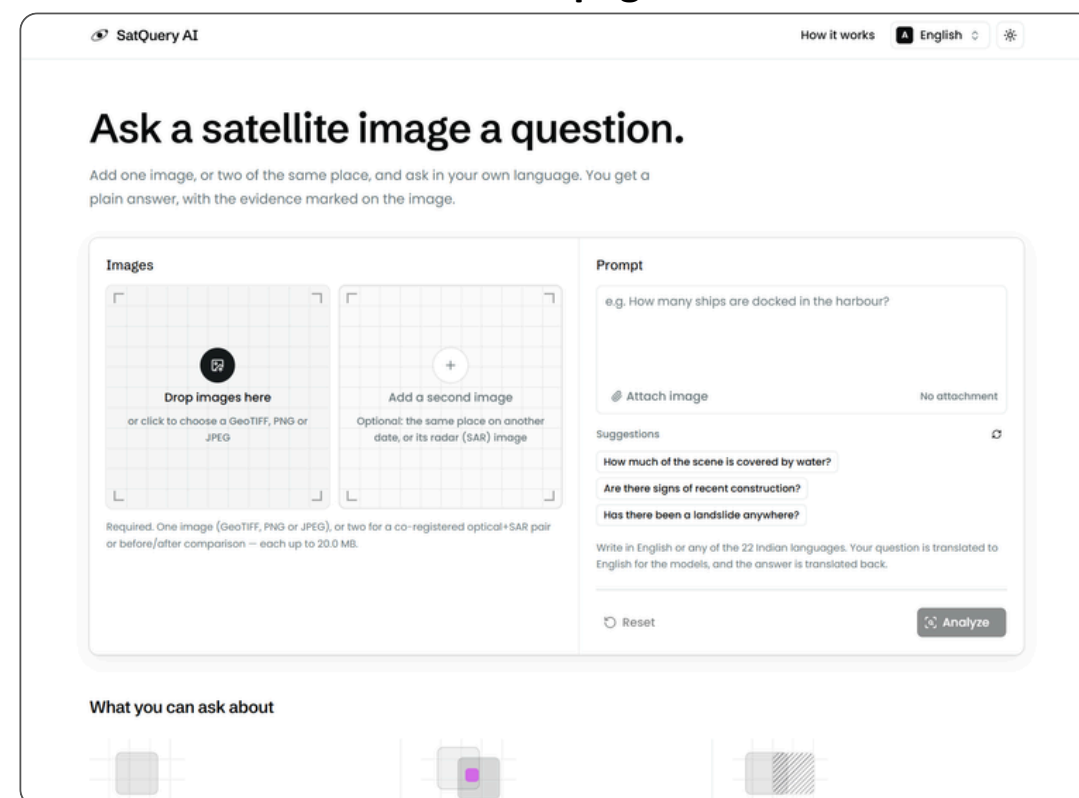


Deployment Architecture

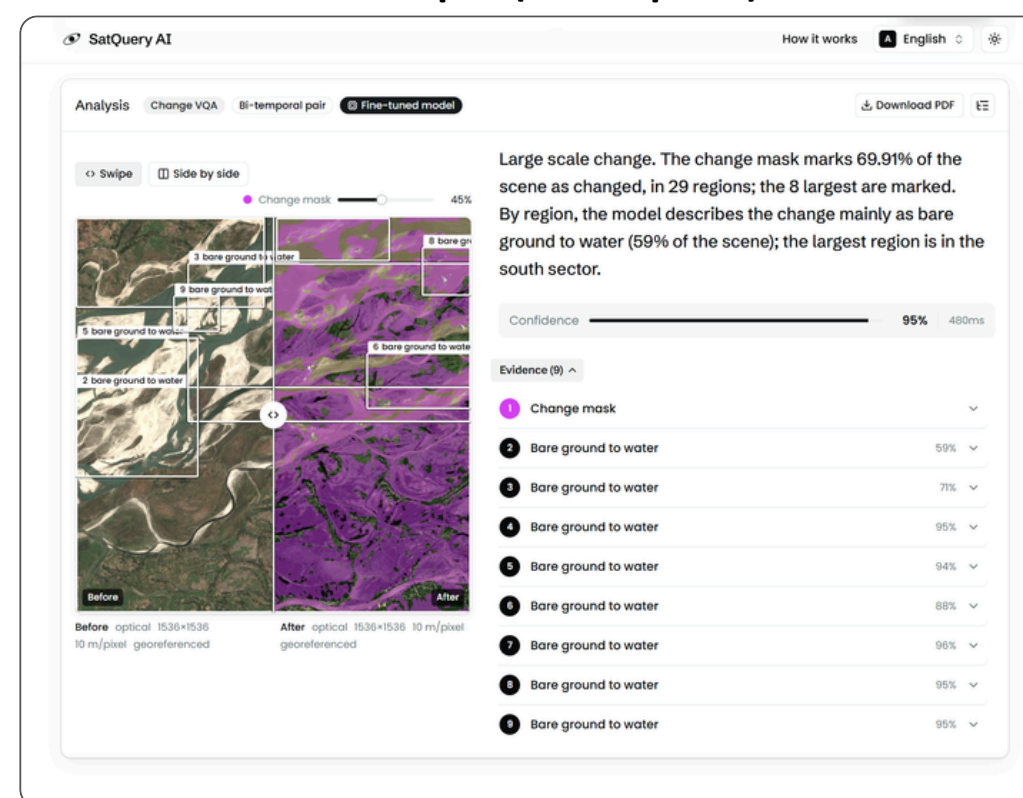


Model Architecture

Output (Bi-temporal)



Home page



Fine Tune Specialists:

- VQA: ConvNeXt-Tiny encoder with question cross-attention; fixed answers (yes/no, land cover, counts). Training on RSVQA-LR.
- Grounding: GroundingDINO + 5 re-rankers. 68.2% accuracy at IoU 0.5 on 16,146 VRSBench queries (GeoChat 57.4%, GeoGround 66.0%); 73.7% on DIOR-RSVG test.
- Change-VQA: Siamese ResNet-34. 81–83% answer accuracy, change-mask F1 0.88 on LEVIR-CD.
- Optical–SAR fusion: ResNet-50 dual encoder. 84.7% SAR→optical matching. Flood-water estimate within 5.7 points of hand-drawn labels.

How it works?

1. Ingest and validate. Reads GeoTIFF, JPEG or PNG. Tells optical from SAR, and reads the ground resolution and georeferencing.
2. Understand the query. IndicTans2 translates the input into English (22 Languages Supported).
3. Route by fixed rules. The input type and the question's wording pick the task. The same input always gets the same task, and no LLM guesses.
4. Check the resolution first. Imagery only reaches a fine-tuned model if its resolution is close to what that model was trained on. Otherwise the result carries a warning.
5. Answer with evidence. Each answer comes with bounding boxes, change masks, areas in km² and a confidence score.
6. Deliver. The answer is translated back to the user's language, with a full execution trace, on the interactive dashboard and as a PDF report.

Technologies Used:

- Frontend: React 19 + Vite, TanStack Router, shadcn/ui, Tailwind CSS. Hosted on Cloudflare.
- Backend: FastAPI (Python), with the specialist models served in-process on a Hugging Face Space.
- ML: PyTorch, Hugging Face Transformers, GroundingDINO, ConvNeXt-Tiny, ResNet-34/50.
- Multilingual: IndicTrans2, for 22 Indian languages ↔ English.
- Fallback vision-language model (VLM): Gemma, through OpenRouter.
- MLOps: GitHub Actions for CI/CD. Model weights on the Hugging Face Hub.
- Reports: PDF export typeset with Typst.



Feasibility

- **Portable Deployment:** Runs on a Hugging Face Space; portable to on-prem GPUs.
- **Advanced Raster Engine:** rasterio reads 4-band 16-bit Cartosat and RISAT SAR.
- **Exact-Match Formatting:** Answers normalized for RSVQA/CDVQA scoring.



Future Strategies

- **Scaling Training Data:** Train on BigEarthNet-MM to boost model accuracy.
- **Multi-Hop Reasoning:** Chain multiple specialists sequentially via the controller rather than single-routing.
- **Confidence Calibration:** Fit task-specific curves on validation data to ensure reliable uncertainty estimates.



Challenges

- **Modality Generalization):** The current VQA models will currently struggle to accurately interpret SAR data without separate, specialized retraining.
- **Resolution Loss:** Downscaling massive Cartosat scenes erases critical small-object details.
- **Grounding accuracy stagnant:** Grounding accuracy plateaued at 67% using Grounding DINO, prompting a shift to rerankers.

IMPACT AND BENEFITS

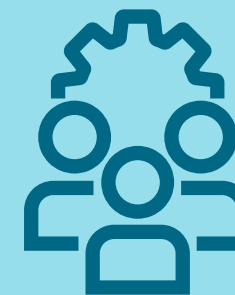
- Dynamically routes complex queries to specialized models via intelligent **Agentic Controller**.
- Generates a fully auditable execution trace for ISRO/SAC evaluators, logging all tasks, tools, and parameters.
- Processes Cartosat and RISAT SAR GeoTIFFs to deliver visually grounded bounding boxes and delivering visually grounded bounding boxes, precise hectare calculations, and model confidence scores.

Strategic Edge (ISRO & Govt.)



- Produces a strict execution trace logging every model, parameter, and compatibility check.
- Deploys specialist tools that calculate exact spatial areas in hectares.
- Democratizes satellite intelligence by allowing non-technical officials to query satellite scenes using plain-English prompts.

Social & Humanitarian (Disaster Relief)



- Maps floods in all weather using SAR speckle removal and automated thresholding.
- Fuse bi-temporal optical and SAR data to detect changes through cloud cover.
- Visualizes impact instantly via interactive before/after change sliders.

Environmental Resilience



- Calculates NDWI and NDVI indices utilizing near-infrared optical bands.
- Automates bi-temporal change detection via optical differencing and SAR log-ratio mapping.
- Delivers precise metrics on the total scene percentage affected by environmental shifts.

KEY REFERENCES

- **ISRO / NRSC — Bhuvan Geoportal**

India's geospatial platform for satellite-image visualisation, thematic mapping and Earth-observation applications :

<https://bhuvan.nrsc.gov.in/ngmaps>

- **RSVQA — Visual Question Answering for Remote Sensing Data**

Foundational research demonstrating how natural-language questions can be used to extract information from remote-sensing imagery

<https://arxiv.org/abs/2003.07333>

- **Large Vision-Language Models for Remote Sensing VQA — 2024**

Explores generative vision-language models for answering open-ended questions about satellite imagery

<https://arxiv.org/abs/2003.07333>

- **MLaVQA — Remote Sensing VQA with Large Language Models — 2026**

Recent research demonstrating multimodal language-model approaches for interpreting complex remote-sensing queries.

<https://www.sciencedirect.com/science/article/abs/pii/S002002552600229X>

- *VRSBench (2024) — the actual benchmark dataset your VQA, captioning, and grounding specialists are trained/evaluated on. Citing it shows judges your model choices aren't arbitrary.* <https://arxiv.org/abs/2406.12384>

- *BigEarthNet-MM (2021) — the paired Sentinel-1/Sentinel-2 (SAR+optical) benchmark archive; this is the direct predecessor/basis for your optical-SAR fusion training data.* <https://arxiv.org/abs/2105.07921>